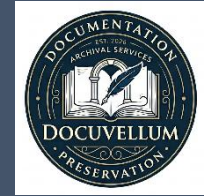


DocuVellum



OBM MANUAL

ABOUT US



ABOUT THIS MANUAL

This manual provides essential information for the safe and effective use of the Outboard Motor (OBM). It outlines the key components of the OBM system—including navigation, propulsion, communications, sensors, and power management—along with operational guidelines, routine maintenance schedules, basic repair procedures, and troubleshooting tips. Designed for use in marine environments ranging from research to security, the OBM operates autonomously or via remote control, supporting a variety of payloads and mission profiles. This manual serves as a practical reference for operators, technicians, and engineers, ensuring reliable performance, system longevity, and safe operation in the field.

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1 GENERAL INFORMATION

1.1 SAFETY INFORMATION BEFORE OPERATING THE OUTBOARD MOTOR

Before mounting or operating the outboard motor, **read this entire manual carefully**. Doing so will help you understand the motor and how to operate it safely.

- **Read all manuals and labels** before operating the motor. Make sure you fully understand each instruction.
- **Do not overpower the boat** with this outboard motor. Using a motor with power exceeding the boat's rated capacity may result in loss of control.
 - The rated power of the outboard motor should be equal to or less than the boat's rated horsepower capacity.
 - If the boat's horsepower rating is unknown, consult the boat manufacturer or dealer.
- **Do not modify the outboard motor.** Any unauthorized modifications may make the motor unsafe or unfit for use.
- **Never operate the motor under the influence of alcohol or drugs.** Approximately 50% of all boating fatalities involve intoxication.
- **Carry a U.S. Coast Guard-approved personal flotation device (PFD) for every person on board.**
 - It is strongly recommended that everyone wears a PFD while boating.
 - At a minimum, children and non-swimmers should always wear a PFD.
 - Everyone should wear a PFD in hazardous boating conditions.
- **Know and follow all local marine laws and regulations.**
- **Stay updated on weather conditions.**

- Check weather forecasts before setting out.
- Avoid boating during hazardous weather.

- **Leave a Float Plan.**

- Inform a responsible person about your boating plans and expected return time.
- Don't forget to cancel the Float Plan upon your safe return.

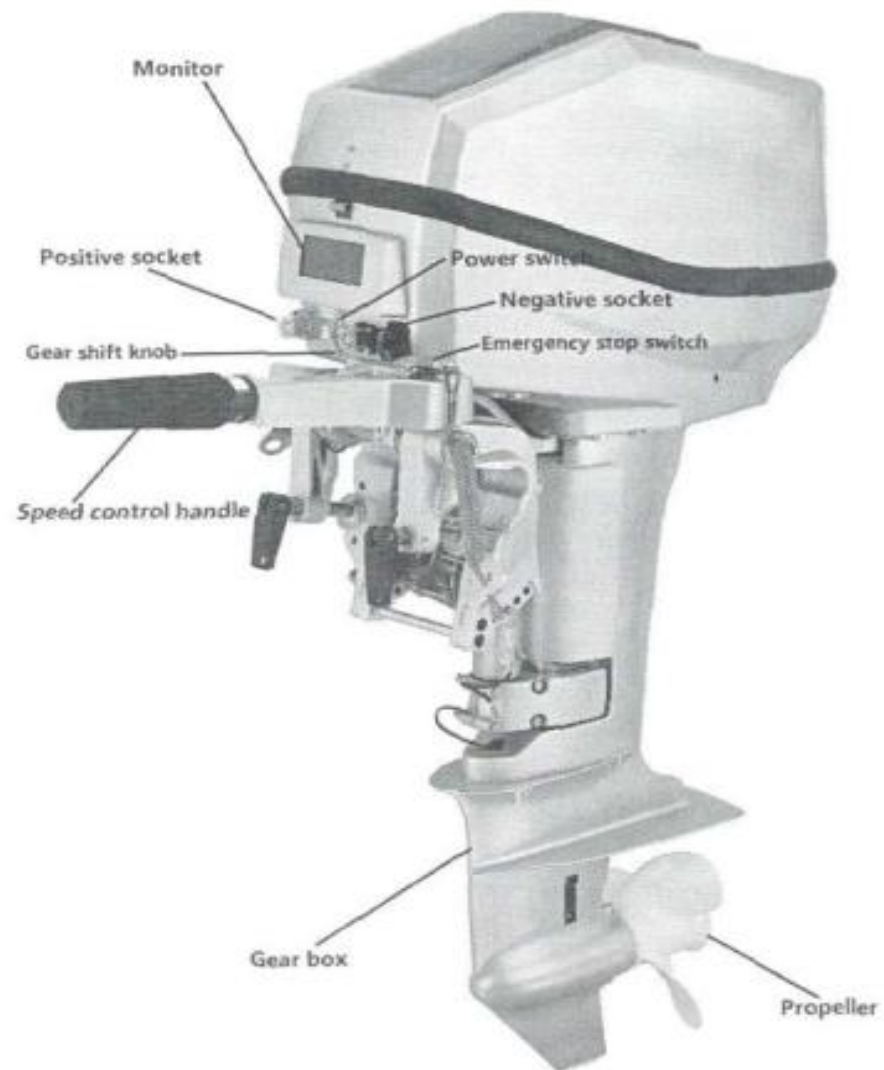
- **Use common sense and good judgment when boating.**

- Know your abilities, and understand how your boat handles in various conditions.
- Operate within your limits and within the boat's limits.
- Always operate at safe speeds and remain alert for obstacles and other boats.

- **Watch carefully for swimmers while the engine is running.**

- Stay clear of swimming areas.
- If someone is in the water near your boat, **shut off the engine immediately**.

1.2 MAIN COMPONENTS



1.3 MAIN TECHNIQUE PARAMETER

PARAMETER	E20S/E20L	PARAMETER	E20S/E20L
Motor	Permanent magnet synchronous motor	Timing	Continuously variable
Rated voltage (V)	DC96	Forward/Reverse	Key switch
Rated power (KW/min-1)	15/5800	Tilting	manual
Recommended boat size	4.2–7m	Propeller (inch)	9¼ * 12-J
Adapted battery voltage (V)	84–108	Adapted transom height short shaft (cm)	42
Max. current (A)	140	Adapted transom height long shaft (cm)	55
Cruise function	yes	Gear oil (ml)	250
Protection class	IP55	Recommended gear oil	Hypoid gear oil SAE 90#
Recommended battery capacity	≥ 300Ah, current ≥ 300A	Net Weight (Short shaft)	52
Power cord interface	Anderson Double-hole straight plug	Net Weight (Long shaft)	54
Indicator	yes		

1.4 PROPELLER SELECTION

The performance of your outboard motor will be critically affected by your choice of propeller, as an incorrect choice could adversely affect performance and could also seriously damage the motor. Engine speed depends on the propeller size and boat load. If engine speed is too high or too low for good engine performance, this will have an adverse effect on the engine the outboard motors are equipped with corresponding series of propellers.

NOTE:

At full throttle and under a maximum boat load, the engine's RPS should be within the upper half of the full throttle operating range.

If operating under conditions which allow the engine's rpm to rise above the maximum recommended range (such as light boat loads), reduce the throttle setting to maintain the rpm in the proper operating range.

1. Propeller diameter (in inches)
2. Propeller pitch (in inches)
3. Type of propeller (propeller mark)

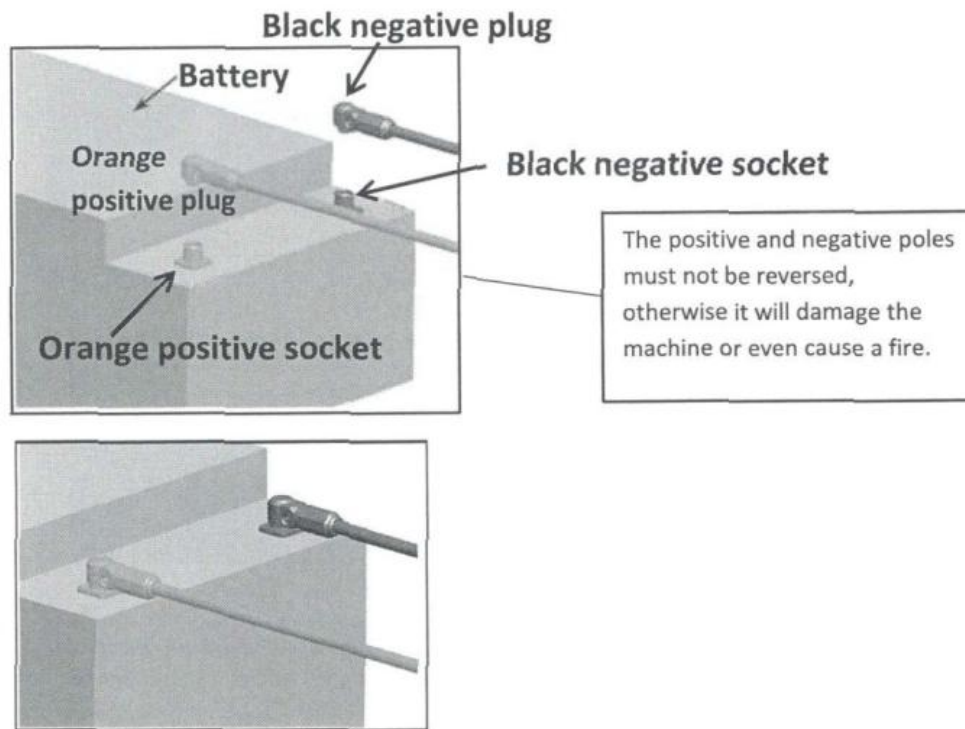


2 OPERATIONS AND OTHER FUNCTIONS

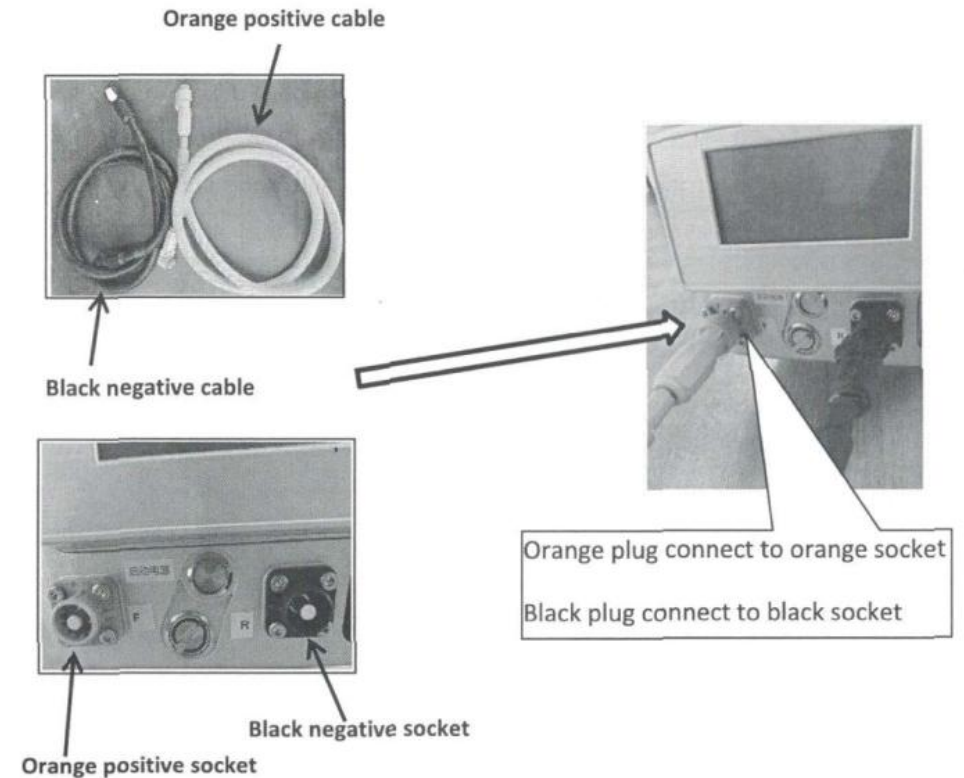
2.1 PRE-OPERATION CHECKS

2.1.1 BATTERY AND LIFESAVING DEVICE

1. Prepare the batteries recommended in the above parameter table
2. Prepare enough life jackets suitable for adults and children
3. Prepare at least two paddles in case the boat cannot move if the motor broke down
4. The orange or red wire is connected to the positive pole (+) of the battery, and the black wire is connected to the negative pole (-) of the battery, as shown below.



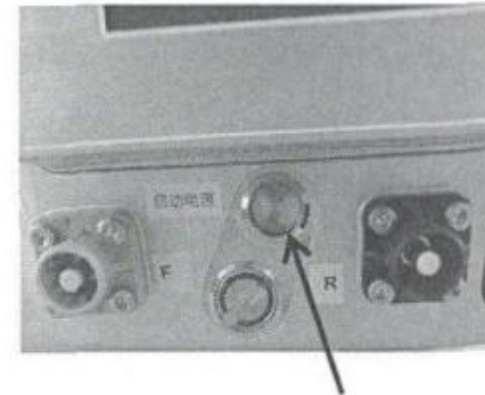
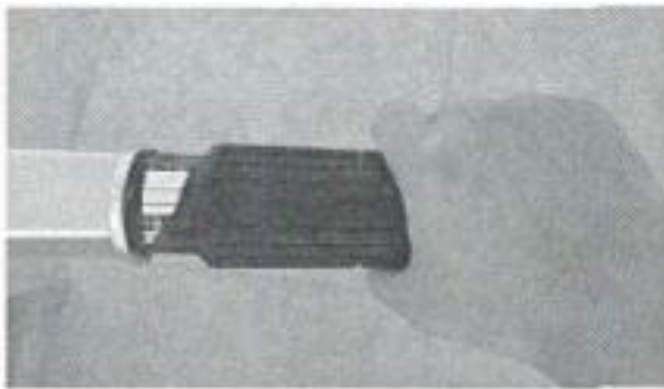
5. Connect the power cable, as shown in the BELOW picture: Insert the orange positive wire into the orange socket. Plug the black negative cable into the black socket



2.2 CONTROL PARTS

2.2.1 STEERING HANDLE ASSEMBLY

1. **Steering handle:** Move the steering handle change the steering direction.
2. **Power control switch:** Press the switch to turn on the power, press the switch again, the switch bounces, and the power is cut off.
3. **The emergency stop** cable with a clamp, please insert the clamp on the switch when the motor in use. In case of emergency, removed the cable, the motor stops running.
4. Change the speed by rotating the handle,



Power control switch



Emergency stop switch

5. **Forward/Reverse:** Rotate the knob, the arrow pointing F, it's FORWARD; the arrow pointing R, it's REVERSE.

2.2.2 STEERING FRICTION ADJUSTMENT SCREW

The friction device provides resistance to the steering movement. This is adjusted for operator's preference. An adjusting screw/bolt is located on the swivel bracket.

Adjustment

- **Resistance:** Screw/bolt
- **Increased:** Turn Clockwise
- **Decreased:** Turn counterclockwise



Forward/Reverse switch

WARNING

Do not overtighten the friction bolts. If the resistance is too great, driving will be difficult, which may cause an accident

2.2.3 TRIM ANGLE ADJUSTING ROD

The position of the trim angle adjusting rod determines the minimum trim angle of the outboard motor in relation to the transom.



2.2.4 TILT LOCK MECHANISM

The tilt-lock mechanism is used to provide reverse thrust from the propeller lifting the outboard motor when reversing. To lock it, set the tilt-lock lever in the lock position. To release it, place the tilt-lock lever in the tilt position.



2.2.5 TILT SUPPORT BAR

The tilt support bar keeps the outboard motor in the tilted-up position.



2.2.6 TOP COWLING LOCK LEVER

To remove the engine top cowling, turn the lock lever. Then lift off the cowling. When replacing the cowling, check to be sure it fits properly in the rubber seal. Then lock the cowling again by moving the lever upward. **Top Cowling lock lever (1)**



2.3 INSTALLATION

CAUTION

Incorrect engine height or obstructions to smooth water flow (such as the design or condition of the boat or accessories such as transom ladders/depth finder transducers) can create airborne water spray while the boat is cruising. Severe engine damage may result if the

NOTE

During water testing check the buoyancy of the boat, at rest, with its maximum load. Check that the static water level on the exhaust housing is low enough to prevent water entry into the powerhead, when water rises due to waves when the outboard is not running.

2.3.1 MOUNTING THE OUTBOARD MOTOR

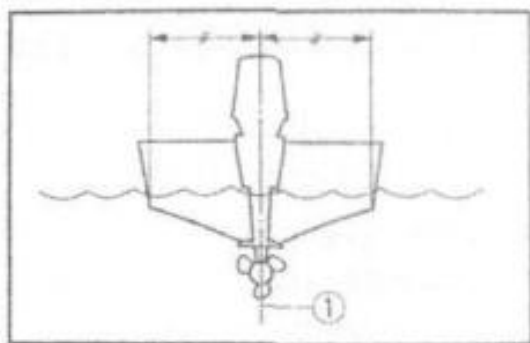
WARNING

Overpowering a boat may cause severe instability. Do not install an outboard motor with more horsepower than the maximum rating on the capacity plate of the boat. If the boat does not have a capacity plate, consult the boat manufacturer.;

The information presented in this section is intended as reference only. It is not possible to provide complete instructions for every possible

Mount the outboard motor on the center line (**keel line**) of the boat, and ensure that the boat itself is well balanced. Otherwise, the boat will be hard to steer. For boats without a keel or which are asymmetrical, consult your dealer.

(1) Center line (keel line)



Mounting Height

To run your boat at optimum efficiency, the water-resistance (drag) of the boat and outboard motor must be made as little as possible. The mounting-height of the outboard motor greatly affects the water-resistance.

If the mounting is too high, cavitation tends to occur, thus reducing the propulsion, and if the propeller tips cut the air, the engine speed will rise abnormally and cause the engine to overheat. If the mounting-height is too low, the water-resistance will increase and thereby reduce engine efficiency.

WARNING

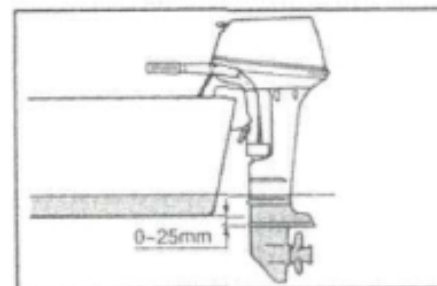
Improper mounting of the outboard motor could result in hazardous conditions such as poor handling, loss of control, or fire hazards.

Observe the following:

For permanently mounted models your dealer or other person experienced in proper rigging should mount the motor. If you are mounting the motor yourself, you should be trained by an experienced person.

For portable models, your dealer or other person experienced in proper outboard motor mounting should show you how to mount your motor.

Mount the engine so that the anti-cavitation plate is between the bottom of the boat and a level 25mm (1 in.) below it.



NOTE

The optimum mounting height of the outboard motor is affected by the boat/motor combination and the desired use. Test runs at different heights can help determine the optimum mounting height.

Refer to the section TRIMMING OUTBOARD MOTOR for instructions on setting the trim angle of the outboard

2.3.2 CLAMPING THE OUTBOARD MOTOR

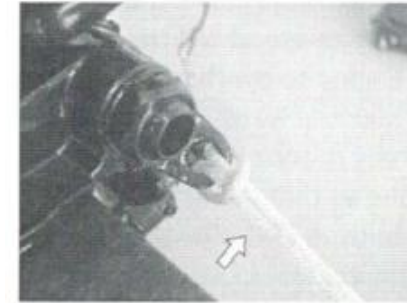
Place the outboard on the transom so that it is positioned as close to the center as possible. Tighten the transom clamp screws evenly and securely. Check the clamp-screws for tightness occasionally during operation of the motor as they can work loose due to engine vibration.



WARNING

Loose clamp screws could allow the motor to move on the transom or fall off the transom. This could cause loss of control and serious injury. Make sure the transom screws are tightened securely. Occasionally check the screws for tightness during operation.

An engine restraint cable or chain should be used. Attach one end to the engine restraint cable attachment point and the other to a secure mounting point on the boat. Otherwise, the engine could be completely lost if it accidentally falls off the transom.



2.3.3 PRE-OPERATION CHECKS

WARNING

If any item in the pre-operation check is not working properly, have it inspected and repaired before operating the outboard motor. Otherwise, an accident could occur.

POWER AND POWER CORD

- Check that the wires are not broken or short-circuited.
- Connect the power supply and make sure that the positive and negative poles of the battery are connected correctly.
- Press the power control switch to check if there is enough power to drive

CONTROLS

- Check throttle, shift and steering for proper operation before starting the engine.
- The controls should work smoothly, without binding or unusual free play.
- Look for loose or damaged connections.
- Check operation of the starter and stop switches when the outboard motor is in the water.

2.3.4 STARTING

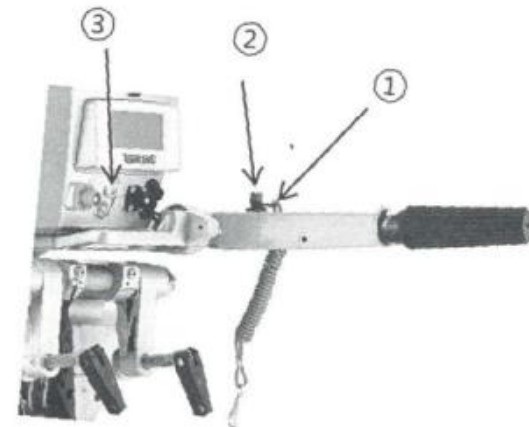
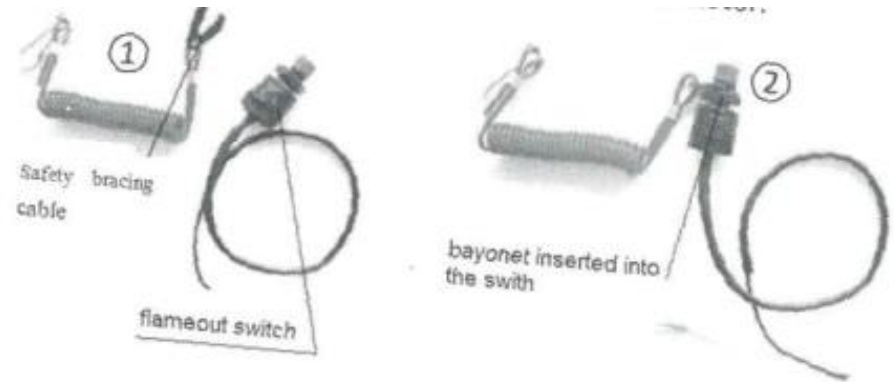
WARNING

Before starting the engine, make sure that the boat is tightly moored and that you can steer clear of any obstructions. Be sure there are no swimmers in the water near you.

This electric outboard motor is with water-cooling system, so it cannot be started out of water. Otherwise, it will get damaged due to overheating.

Please mount the outboard motor correctly and ensure that the designated part of the outboard motor is placed in water before starting the outboard engine.

1. Inserted the safety bracing Cable (1) into the Switch (2).
2. Press the power Control switch (3) to turn ON the motor.



2.3.5 SPEED ADJUSTING

Slowly turn the throttle handle to adjust the speed,

2.3.6 SHIFTING

WARNING

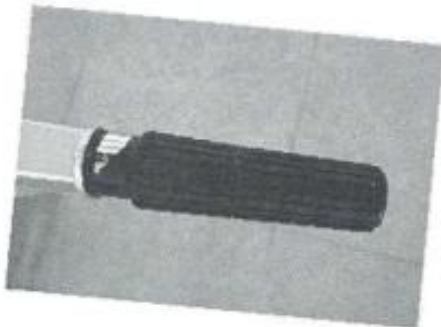
Before shifting make sure there are no swimmers/obstacles in the water near you.

FORWARD/; REVERSE

- 1) Place the throttle control grip in the fully closed position and then shift gears.
- 2) Make Sure the tilt-lock lever in the lock position,

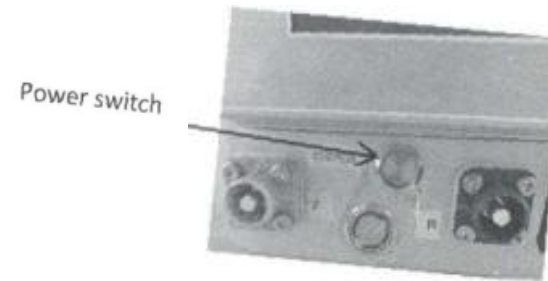
WARNING

When operating in reverse, go slowly. Do not open the throttle more than half. Otherwise, the boat may become unstable, which could result in loss of control and an accident



2.3.7 STOPPING ENGINE

1. Place the throttle control grip in the fully closed position, the machine stops running, press the power control switch, the switch pops up, then the power has been cut off.
2. After stopping the engine, disconnect the power cord.



2.3.8 TRIMMING OUTBOARD MOTOR

The trim angle of the outboard motor helps determine the position of the bow of the boat in the water. The correct trim angle will help improve performance and fuel economy while reducing strain on the engine. The correct trim angle depends upon the combination of boat, engine and propeller. Correct trim is also affected by variables such as load in the boat, water conditions and running speed.

WARNING

Excessive trim for the operating conditions (either trim up or trim down) can cause boat instability and can make steering the boat more difficult. This increases the possibility of an accident. If the boat begins to feel unstable or is hard to steer, slow down and/or readjust the trim angle.



NOTE

- Refer to the section ADJUSTING TRIM ANGLE for instructions on usage.

2.3.9 ADJUSTING TRIM ANGLE

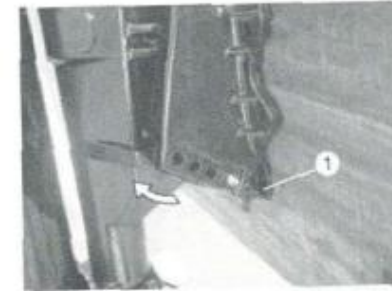
Manual tilt model

There are 4 or 5 holes provided in the clamp bracket to adjust the outboard motor trim angle.

1. Stop the engine and cut off the power.
2. Remove the trim angle and adjusting rod (1) from the clamp bracket while tilting the motor up slightly.

NOTE

The outboard motor trim angle can be changed approximately 4 degrees by shifting the trim adjusting-rod one hole.



3. Reposition the rod in the desired hole. To raise the bow(trim-in), move the rod away from the transom. To lower the bow (trim out), move the rod towards the transom.
4. Make test runs with the trim set to different angles to find the position that works best for your boat and operating conditions.

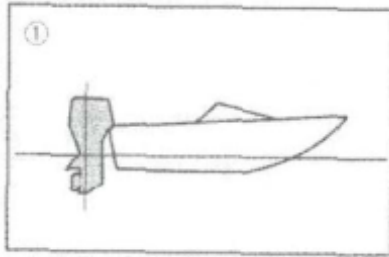
WARNING

- Stop the engine before adjusting the trim angle.
- Take care to avoid being pinched when removing or installing the rod.
- Be cautious when trying a trim position for the first time. Increase Speed gradually and watch for any signs of instability or control problems. An improper tilt angle can cause the boat to lose control.

2.3.10 TRIM ANGLE SETTINGS AND BOAT HANDLING

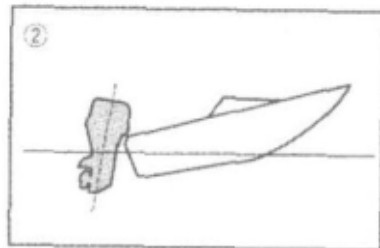
OPTIMUM ANGLE (LEVEL KEEL LINE)

When the ship is traveling at high speed, the bow slightly upward can reduce drag and improve stability and efficiency. This is generally when the keel line of the boat is up about 3 to 5 degrees. With the bow up, the boat may have a greater tendency to steer to one side or the other. Compensate for this as you steer. The trim tab can also be adjusted to help offset this effect.



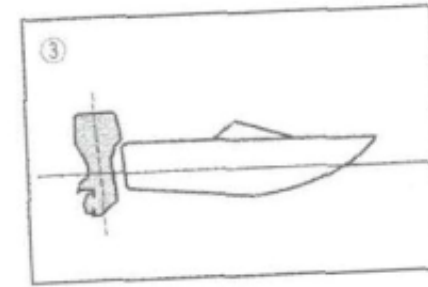
BOW UP

Too much trim-out puts the bow of the boat high in the water. Performance and decreased because the hull of the boat is pushing the water and there is more air drag. Excessive trim-out can also cause the propeller to ventilate, which reduces performance further, and the boat may porpoise (hop in the water), which could throw the operator and passengers overboard.



BOW DOWN

When the bow sinks, it is easier to accelerate from a standstill to driving fast. Too much trim-in causes the boat to plow through the water, decreasing fuel economy and making it hard to increase speed. Operating with excessive trim-in at higher speeds also makes the boat unstable. Resistance at the bow is greatly increased, heightening the danger of bow steering and making operation difficult and dangerous.



NOTE

Depending on the type of boat, the outboard motor trim angle may have little effect on the trim of the boat when operating

TILTING UP/DOWN

If the engine will be stopped for some time, or if the boat is moored in shallows, the engine should be tilted up to protect the propeller and casing from damage by collision with obstructions, and also to reduce salt corrosion.



WARNING

Be sure all people are clear of the outboard motor when adjusting the tilt angle, also be careful not to pinch any body parts between the drive unit and engine bracket.

CAUTION

Before tilting the motor, follow the procedures under **STOPPING ENGINE**. Never tilt the motor while the engine is running. Severe damage from overheating can result. Do not tilt up the engine by pushing the steering handle as this could break the handle.

PROCEDURE FOR TILTING UP

MANUAL TILT MODEL

1. Stop engine
2. Place the tilt lock lever in the release position.



3. Hold the rear of the top cowling with one hand and fully tilt the engine up.
4. The tilt support bar turns to the locked position automatically.



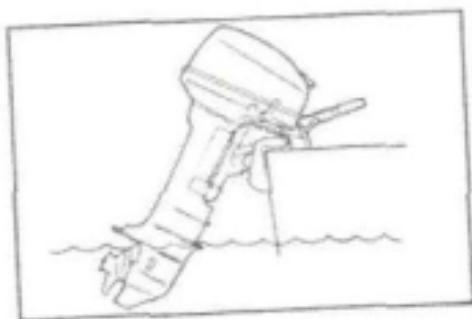
PROCEDURE FOR TILTING DOWN



1. Stop engine
2. Place the tilt lock lever in the lock position
3. Slightly tilt up the engine until the tilt support bar is released automatically.
4. Tilt down the engine.

CRUISING IN SHALLOW WATER

The outboard motor can be tilted up partially to allow operation in shallow water.



WARNING

1. Run the boat at the lowest possible speed when using the shallow water cruising
2. system. The tilt-lock mechanism does not work while the shallow water cruising system is being used. Hitting an underwater obstacle could cause the engine to lift out of the water, resulting in loss of control.
3. Do not rotate the engine 180° and operate the boat in reverse. Place the gear-shift in
4. the Reverse position to operate the boat in reverse.
5. Use extra care when operating in reverse. Too much reverse thrust can cause the
6. engine to lift out of the water, increasing the chance of accident and personal injury.

PROCEDURE

1. Place the throttle control grip in the fully closed position, the machine stops running, press the power control button to cut off the power.
2. Place the tilt lock lever in the release position.
3. Slightly tilt up the engine. The tilt-support bar will lock automatically, supporting the engine in a partially raised position.



NOTE

This motor has 2 positions for shallow water cruising.

RETURNING TO ORIGINAL POSITION

1. Place the tilt lock lever in the lock position.
2. Slightly tilt up the engine until the tilt support bar automatically returns to the free position.
3. Then, slowly lower the engine to the normal position.



3 MAINTENANCE

The motor should be trailered and stored in the normal running position. If there is insufficient road clearance in this position, then trailer the motor in the tilt position using a motor support device such as a transom saver bar.

WARNING

Never get under the lower unit while it is tilted, even if a motor support bar is used. Severe injury could occur if the outboard accidentally falls.

NOTE

Place a towel or the like under the outboard motor to protect it from damage.

CAUTION

Do not use the tilt support lever/knob when trailering the boat. The outboard motor could shake loose from the tilt support and fall. If the motor cannot be trailered in the down position, use an additional support device to secure it in the up position.



3.1 STORING OUTBOARD MOTOR

When storing your outboard motor for prolonged periods of time (2 months or longer), several important procedures must be performed to prevent excessive damage. It is advisable to have your outboard motor serviced by an authorized dealer prior to storage. However, the following procedures can be performed by the owner with a minimum tool.

1. Cut off the power and remove the battery.
2. Wash the motor body using fresh water (Refer to Motor EXTERIOR for details).

CAUTION

Before the water is completely drained, do not put the outboard motor on its side, otherwise the water may flow into the motor and controller and cause damage and corrosion. Store the outboard motor in a dry, well-ventilated place, not in direct sunlight.

3.2 PERIODIC MAINTENANCE

REPLACEMENT PARTS

If replacement parts are necessary, use only genuine parts or equivalents of the same type and of equivalent strength and materials. Any part of inferior quality may malfunction, and the resulting loss of control could endanger the operator and passengers.

MAINTENANCE CHART

Frequency of maintenance operations may be adjusted according to the operating conditions, but the following table gives general guidelines. Refer to the sections in this chapter for explanations of each owner-specific action

The mark (#) indicates the check-ups which you may carry out by yourself.
The mark (0) indicates works to be carried out by your dealer.

WARNING

Be sure to turn off the engine when you perform maintenance unless otherwise specified. If the owner is not familiar with machine servicing, this work should be done by a dealer or other qualified mechanic.

Item	Actions	10 Hours (1 month)	50 Hours (3 months)	100 Hours (6 months)	200 Hours (1 year)
Greasing points	Greasing			●	
Gear oil	Change	●	●	●	
External anode	Inspection/replacement		●/○	●/○	
Propeller, cotter pin	Inspection/replacement		●	●	
Top cowling	Inspection				●
Engine	Inspection			●	
Electric wire	Inspection	●		●	

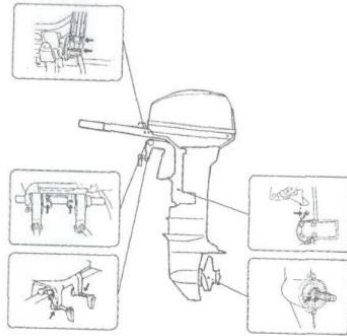
NOTE

When operating in salt water, turbid or muddy water, the engine should be flushed with clean water after each use.

3.3 GREASING

Grease A (Water resistant grease)

Grease D (Corrosion resistant grease) *1



3.4 INSPECTING IDLING SPEED

Do not touch or remove electrical parts when starting or during operation.
Keep hands, hair and clothes away from rotating parts while engine running.

3.5 CHECKING PROPELLER

WARNING

You could be seriously injured if the engine accidentally starts while you are near the propeller before inspecting removing or installing the propeller, unplug the power cord. Do not hold the propeller by your hand when loosening or tightening the nuts for propeller. Put a wood block between the depressed plate and the propeller to prevent the propeller from turning.

PROPELLER CHECKING POINT

- Check each of the propeller blades for wear, erosion from cavitation or ventilation, or other damage.
- Check the propeller shaft for damage.
- Check the shear pin for wear or damage.
- Check for fish line winding around the propeller shaft.
- Check the propeller shaft oil seal for damage.

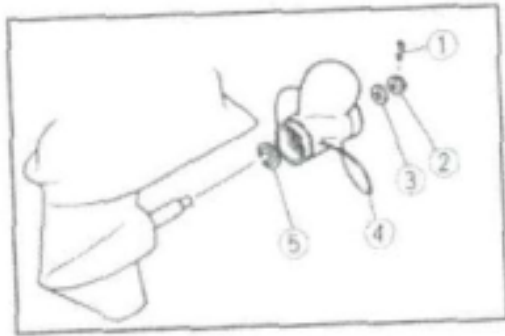


REMOVING THE PROPELLER

Straighten the cotter-pin (1) and pull it out using a pair of pliers.

Remove the propeller nut (2) and washer (3).

Remove the propeller (4) and thrust washer (5).

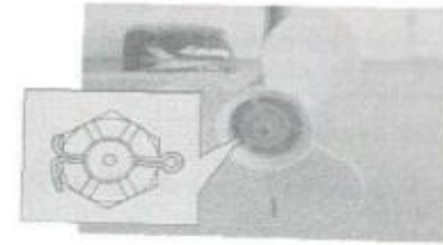


INSTALLING THE PROPELLER

CAUTION

Be sure to install the thrust washer before installing the propeller, otherwise the lower case and propeller boss may be damaged.

Be sure to use a new cotter pin and bend the ends over securely. Otherwise, the propeller could come off during operation and be lost.



1. Apply grease or a corrosion resistant grease to the propeller-shaft.
2. Install the thrust washer and propeller on the propeller-shaft.
3. Install the washer and tighten the propeller nut to the specified torque.
4. Align the propeller nut with the propeller shaft hole. Insert a new cotter pin in the hole and bend the cotter pin ends.

NOTE

After tightening to the specified torque, if the propeller nut is not aligned with the propeller shaft hole, further tighten the nut to align with the shaft hole.

3.6 CHANGING GEAR OIL

WARNING

Be sure the outboard motor is securely fastened to the transom or a stable stand. You could be severely injured if the outboard motor falls on you.

Never get under the lower unit while it is tilted, even when the tilt support lever or knob is locked. Severe injury could occur if the outboard motor accidentally falls.

1. Tilt the outboard motor so that gear oil drain screw is at the lowest point possible.
2. Place a suitable container under the gear case.
3. Remove the gear oil drain screw (1),
4. Remove the oil level plug (2) to allow the oil to drain completely.
5. With the outboard motor in a vertical position, and using a flexible or pressurized filling device, inject the gear oil drain screw hole.
6. When the oil begins to flow out of the oil level plug hole, insert and tighten the oil level plug.
7. Insert and tighten the gear oil drain screw.



CAUTION

Inspect the used oil after it has been drained. If the oil is milky, water is getting into the gear case which can cause gear damage. The seals need to be changed.

3.7 INSPECTING AND REPLACING ANODE

The outboard motors are protected from corrosion by sacrificial anodes. Inspect the external anodes periodically. Remove scales from the surfaces of the anodes.



CAUTION

Do not paint anodes, as this would render them ineffective.

NOTE

Inspect ground leads attached to external anodes on equipped models.

3.8 CLEANING THE OUTBOARD MOTOR

After use, wash the exterior of the outboard with fresh water.

3.9 CHECKING PAINTED SURFACE OF MOTOR

Check the motor for scratches, nicks, or flaking paint. Areas with damaged paint are more likely to corrode. If necessary, clean and paint the areas.

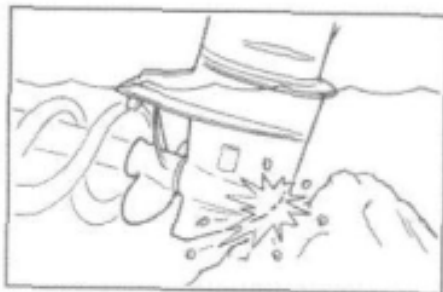
4 TEMPORARY ACTION IN EMERGENCY

4.1 IMPACT DAMAGE

WARNING

The outboard motor can be seriously damaged by a collision while operating or trailering. Damage could make the outboard motor unsafe to operate,

1. If the outboard motor hits any object in the water, follow the procedure below.
2. Stop the engine immediately.
3. Inspect the control system and all components for damage. Also, inspect the boat for damage.
4. Whether damage is found or not, return to the nearest harbor slowly and carefully.
5. Have a TOMKING dealer inspect the outboard motor before operating it again.



4.2 TREATMENT OF SUBMERGED MOTOR

If the outboard motor falls into the water, please disconnect the power cord immediately, and clean the machine immediately after taking it out from water.

Otherwise, it will be corroded. If you cannot take the outboard motor to a dealer or after-sales service immediately, please follow the procedure below to minimize engine damage.

1. Thoroughly wash mud, salt, seaweed, etc, with clean fresh water.
2. Take the outboard motor to a TOMKING dealer as soon as possible.



CAUTION

Do not attempt to run the motor until it has been completely inspected.